

Lie Groups, Lie Algebras, and Representations

Reading Notes and Exercises

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1 Matrix Lie Groups

1.1 Definitions

Definition 1.1. The **general linear group** over the real numbers, denoted $GL(n, \mathbb{R})$, is the group of all $n \times n$ invertible matrices with real entries. The general linear group over the complex numbers, denoted $GL(n, \mathbb{C})$, is the group of all $n \times n$ invertible matrices with complex entries.

Definition 1.2. Let $M_n(\mathbb{C})$ denote the space of all $n \times n$ matrices with complex entries.

Definition 1.3. Let A_m be a sequence of complex matrices in $M_n(\mathbb{C})$. We say A_m **converges** to a matrix A if each entry of A_m converges to the corresponding entry of A as $m \rightarrow \infty$, i.e. if $(A_m)_{jk}$ converges to A_{jk} for all $1 \leq j, k \leq n$.

Note that Definition 1.3 means we can identify $M_n(\mathbb{C})$ with $\mathbb{C}^{n^2}$, and use the standard notion of convergence in $\mathbb{C}^{n^2}$. All topological properties of a matrix Lie group G will be considered with respect to the topology G inherits as a subset of $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$.

We now consider subgroups of $GL(n, \mathbb{C})$, which are subsets $G \subset GL(n, \mathbb{C})$ such that $Id \in G$, and for all $A, B \in G$ we have $AB \in G$ and $A^{-1} \in G$.

Definition 1.4. A **matrix Lie group** is a subgroup G of $GL(n, \mathbb{C})$ with the property: if A_m is any sequence of matrices in G , and A_m converges to some matrix A , then either A is in G or A is not invertible.

This property means that G is a closed subset of $GL(n, \mathbb{C})$. So this definition is equivalent to saying a matrix Lie group is a closed subgroup of $GL(n, \mathbb{C})$. However, G is not necessarily closed in $M_n(\mathbb{C})$. For this to be true, we require a stronger property: if A_m is any sequence of matrices in G and A_m converges to some matrix A , then $A \in G$.

1.2 Examples

We introduce some important examples of (matrix) Lie groups, including the classical groups.

1.2.1 General and Special Linear Groups

The general linear groups $GL(n, \mathbb{R})$ and $GL(n, \mathbb{C})$, defined in 1.1, are matrix Lie groups.

The special linear groups $SL(n, \mathbb{R})$ and $SL(n, \mathbb{C})$ is the group of $n \times n$ invertible matrices (with real or complex entries) having determinant 1. They are matrix Lie groups.

1.2.2 Unitary and Orthogonal Groups

A $n \times n$ complex matrix A is **unitary** if its column vectors are orthonormal, that is,

$$\sum_{l=1}^n \bar{A}_{lj} A_{lk} = \delta_{jk}$$

where δ_{jk} is the Kronecker delta. This can be rewritten as

$$\sum_{l=1}^n (A^*)_{jl} A_{lk} = \delta_{jk}$$

where A^* is the adjoint of A , defined by $A^* = \bar{A}^t$. A is unitary if and only if $A^* = A^{-1}$.

The **unitary group** $U(n)$ is the collection of all unitary matrices. The **special unitary group** $SU(n)$ is the subgroup of $U(n)$ consisting of unitary matrices with determinant 1. $U(n)$ and $SU(n)$ are matrix Lie groups.

A $n \times n$ real matrix A is **orthogonal** if its column vectors are orthonormal. The equivalent conditions are the same as before, except that when A is real, $A^* = A^t$. A is orthogonal if and only if $A^t = A^{-1}$, if and only if A preserves the inner product on $\mathbb{R}^n$.

The **orthogonal group** $O(n)$ is the collection of all orthogonal matrices. The **special orthogonal group** $SO(n)$ is the set of $n \times n$ orthogonal matrices with determinant 1. Elements of $SO(n)$ are rotations, while the elements of $O(n)$ are either rotations or combinations of rotations and reflections.

Consider the bilinear form $(\cdot, \cdot)$ on $\mathbb{C}^n$ defined by

$$(x, y) = \sum_j x_j y_j$$

Note that this form is not an inner product, since it is symmetric, not conjugate-symmetric. The **complex orthogonal group** $O(n, \mathbb{C})$ is the set of all $n \times n$ complex matrices A that preserve this form. The group $SO(n, \mathbb{C})$ is the set of all matrices in $O(n, \mathbb{C})$ with determinant 1. $O(n, \mathbb{C})$ and $SO(n, \mathbb{C})$ are matrix Lie groups.

1.2.3 Generalized Orthogonal and Lorentz Groups

Consider $\mathbb{R}^{n+k}$, with n and k positive integers. Define a symmetric bilinear form $[\cdot, \cdot]_{n,k}$ on $\mathbb{R}^{n+k}$ by

$$[x, y]_{n,k} = x_1 y_1 + \cdots + x_n y_n - x_{n+1} y_{n+1} - \cdots - x_{n+k} y_{n+k}$$

The **generalized orthogonal group** $O(n, k)$ is the set of $(n+k) \times (n+k)$ real matrices A that preserve this form, i.e. $[Ax, Ay]_{n,k} = [x, y]_{n,k}$. It is a matrix Lie group. In particular, the **Lorentz group** is $O(3, 1)$. $SO(n, k)$ is the subgroup of $O(n, k)$ consisting of elements with determinant 1.

For any $A \in O(n, k)$, we have $\det A = \pm 1$.

1.2.4 Symplectic Groups

Define the skew-symmetric bilinear form ω on $\mathbb{R}^{2n}$ by

$$\omega(x, y) = \sum_{j=1}^n (x_j y_{n+j} - x_{n+j} y_j)$$

The **real symplectic group** $Sp(n, \mathbb{R})$ is the set of all $2n \times 2n$ real matrices A which preserve ω . It is a closed subgroup of $GL(2n, \mathbb{R})$, hence a matrix Lie group. $\det A = \pm 1$ for all $A \in Sp(n, \mathbb{R})$; in fact, $\det A = 1$.

Using the same formula, we can define a bilinear form ω on $\mathbb{C}^{2n}$. Over $\mathbb{C}$, we have the relation $\omega(z, w) = (z, \Omega w)$, where $(\cdot, \cdot)$ is the complex bilinear form from 1.2.2. The **complex symplectic group** $Sp(n, \mathbb{C})$ is the set of all $2n \times 2n$ complex matrices which preserve this form. We can show that for all $A \in Sp(n, \mathbb{C})$, $\det A = \pm 1$, and in fact, $\det A = 1$.

The **compact symplectic group** $Sp(n)$ is defined as

$$Sp(n) = Sp(n, \mathbb{C}) \cap U(2n)$$

That is to say, $Sp(n)$ is the group of $2n \times 2n$ matrices that preserve both the inner product and the bilinear form ω .

1.2.5 The Euclidean and Poincaré Groups

The **Euclidean group** $E(n)$ is the group of all transformations of $\mathbb{R}^n$ that can be expressed as a composition of a translation and an orthogonal linear transformation. Elements of $E(n)$ are written as pairs $\{x, R\}$ with $x \in \mathbb{R}^n$ and $R \in O(n)$, and $\{x, R\}$ acts on $\mathbb{R}^n$ by

$$\{x, R\}y = Ry + x$$

The product operation for $E(n)$ is

$$\{x_1, R_1\}\{x_2, R_2\} = \{x_1 + R_1x_2, R_1R_2\}$$

The inverse of an element of $E(n)$ is

$$\{x, R\}^{-1} = \{-R^{-1}x, R^{-1}\}$$

$E(n)$ is not a subgroup of $GL(n, \mathbb{R})$, but it is isomorphic to a closed subgroup of $GL(n+1, \mathbb{R})$.

The **Poincaré group** $P(n, 1)$ is the group of all transformations of $\mathbb{R}^{n+1}$ of the form

$$T = T_x A$$

with $x \in \mathbb{R}^{n+1}$ and $A \in O(n, 1)$.

1.2.6 The Heisenberg Group

The **Heisenberg group** H is the set of all 3×3 real matrices

$$A = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}$$

where $a, b, c \in \mathbb{R}$. H is a subgroup of $GL(3, \mathbb{R})$.

1.2.7 The Groups $\mathbb{R}^*$, $\mathbb{C}^*$, S^1 , $\mathbb{R}$, and $\mathbb{R}^n$

These groups are not defined as groups of matrices, but can be thought of as such.

- The group $\mathbb{R}^*$ of non-zero real numbers under multiplication is isomorphic to $GL(1, \mathbb{R})$
- The group $\mathbb{C}^*$ of non-zero complex numbers under multiplication is isomorphic to $GL(1, \mathbb{C})$
- The group S^1 of complex numbers with absolute value 1 is isomorphic to $U(1)$
- The group $\mathbb{R}$ under addition is isomorphic to $GL(1, \mathbb{R})^+$, the 1×1 matrices with positive determinant
- The group $\mathbb{R}^n$ under vector addition is isomorphic to the group of diagonal real matrices with positive diagonal entries

1.2.8 The Compact Symplectic Group

We try to understand the structure of $Sp(n)$, and show that it can be understood as the "unitary group over the quaternions".

Define a conjugate-linear map $J : \mathbb{C}^{2n} \rightarrow \mathbb{C}^{2n}$ by

$$J(\alpha, \beta) = (-\bar{\beta}, \bar{\alpha})$$

where $\alpha, \beta \in \mathbb{C}^n$ and $(\alpha, \beta) \in \mathbb{C}^{2n}$. For all $z, w \in \mathbb{C}^{2n}$, we have

$$\omega(z, w) = \langle Jz, w \rangle$$

Furthermore, for all $z, w \in \mathbb{C}^{2n}$, we have

$$\langle Jz, w \rangle = -\langle Jw, z \rangle$$

and

$$J^2 = -I$$

Proposition 1.5. *If $U \in U(2n)$, then $U \in Sp(n)$ if and only if U commutes with J .*

Theorem 1.6. *$U \in Sp(n)$ if and only if there exists an orthonormal basis $u_1, \dots, u_n, v_1, \dots, v_n$ for $\mathbb{C}^{2n}$ such that the following properties hold:*

1. $Ju_j = v_j$
2. For some real numbers $\theta_1, \dots, \theta_n$, we have

$$\begin{aligned} Uu_j &= e^{i\theta_j}u_j \\ Uv_j &= e^{-i\theta_j}v_j \end{aligned}$$

3. We have

$$\begin{aligned} \omega(u_j, u_k) &= \omega(v_j, v_k) = 0 \\ \omega(u_j, v_k) &= \delta_{jk} \end{aligned}$$

Lemma 1.7. *Suppose V is a complex subspace of $\mathbb{C}^{2n}$ that is invariant under the conjugate-linear map J . Then the orthogonal complement $V^\perp$ of V (with respect to the inner product $\langle \cdot, \cdot \rangle$) is also invariant under J . Furthermore, V and $V^\perp$ are orthogonal with respect to ω , i.e.*

$$\omega(z, w) = 0$$

for all $z \in V$ and $w \in V^\perp$.

1.3 Topological Properties

1.3.1 Compactness

Definition 1.8. *A matrix Lie group $G \subset GL(n, \mathbb{C})$ is **compact** if it is compact in the usual topological sense as a subset of $M_n(\mathbb{C}) \cong \mathbb{R}^{2n^2}$.*

By the Heine-Borel theorem, a matrix Lie group G is compact if and only if it is closed and bounded.

$O(n)$, $SO(n)$, $U(n)$, $SU(n)$, and $Sp(n)$ are compact.

1.3.2 Connectedness

Definition 1.9. *A matrix Lie group G is **connected** if for all $A, B \in G$, there exists a continuous path $A(t)$, $a \leq t \leq b$, lying in G with $A(a) = A$ and $A(b) = B$. For any matrix Lie group G , the **identity component** of G , denoted G_0 , is the set of $A \in G$ for which there exists a continuous path $A(t)$, $a \leq t \leq b$, lying in G with $A(a) = I$ and $A(b) = A$.*

This definition is actually called **path connected** in topology. A matrix Lie group is connected if and only if it is path-connected.

To show that a matrix Lie group G is connected, it suffices to show that each $A \in G$ can be connected to the identity by a continuous path lying in G .

Proposition 1.10. *If G is a matrix Lie group, the identity component G_0 of G is a normal subgroup of G .*

G_0 is closed, and hence a matrix Lie group.

Proposition 1.11. *$GL(n, \mathbb{C})$ is connected for all $n \geq 1$.*

Proposition 1.12. *$SL(n, \mathbb{C})$ is connected for all $n \geq 1$.*

Proposition 1.13. *$U(n)$, $SU(n)$, and $SO(n)$ are connected for all $n \geq 1$.*

1.3.3 The Topology of $SO(3)$

Definition 1.14. *The **real projective space** of dimension n , denoted $\mathbb{R}P^n$, is the set of lines through the origin in $\mathbb{R}^{n+1}$. Since each line through the origin intersects the unit sphere exactly twice, we may think of $\mathbb{R}P^n$ as the unit sphere S^n with "antipodal" points u and $-u$ identified.*

Points in $\mathbb{R}P^n$ can be thought of as pairs $\{u, -u\}$ with $u \in S^n$. There is a natural map $\pi : S^n \rightarrow \mathbb{R}P^n$

$$\pi(u) = \{u, -u\}$$

A distance function can be defined on $\mathbb{R}P^n$ by

$$d(\{u, -u\}, \{v, -v\}) = \min(d(u, v), d(u, -v))$$

The space $\mathbb{R}P^n$ is the upper hemisphere in S^n with antipodal points on the equator identified. $\mathbb{R}P^n$ is the closed unit ball $B^n \subset \mathbb{R}^n$, with antipodal points on the boundary of B^n identified.

Proposition 1.15. *There is a continuous bijection between $SO(3)$ and $\mathbb{R}P^3$.*

$\mathbb{R}P^n$ is not simply connected, and hence $SO(3)$ is not simply connected.

1.4 Homomorphisms

Definition 1.16. *Let G and H be matrix Lie groups. A map Φ from G to H is called a **Lie group homomorphism** if*

1. Φ is a group homomorphism, and
2. Φ is continuous.

*If, in addition, Φ is one-to-one and onto and the inverse map Φ^{-1} is continuous, then Φ is called a **Lie group isomorphism**.*

If G and H are matrix Lie groups, and there exists a Lie group isomorphism from G to H , then G and H are isomorphic, $G \cong H$.

The relationship between $SU(2)$ and $SO(3)$: they are not isomorphic, but there exists a Lie group homomorphism $\Phi : SU(2) \rightarrow SO(3)$ that is two-to-one and onto.

Proposition 1.17. *The map $U \mapsto \Phi_U$ is a 2-1 and onto map of $SU(2)$ to $SO(3)$, with kernel equal to $\{I, -I\}$.*

$SU(2)$ is homeomorphic to S^3 . The proposition shows that $SO(3)$ is homeomorphic to $\mathbb{R}P^3$, that is, S^3 with antipodal points identified.

1.5 Lie Groups

The text first gives a very brief discussion of manifolds and smooth manifolds.

Definition 1.18. A **Lie group** is a smooth manifold G which is also a group, and such that the group product $G \times G \rightarrow G$ and the inverse map $G \rightarrow G$ are smooth.

Every matrix Lie group is a Lie group, but the reverse is not true. However, we focus on matrix Lie groups.

Usually, we call a map Φ between two Lie groups a Lie group homomorphism if Φ is a group homomorphism and Φ is smooth.

Every matrix Lie group G is a manifold, so G must be locally path connected, so G is connected if and only if it is path connected.

1.6 Exercises

Exercise 1.7 Using Theorem 1.6, show that $Sp(n)$ is connected and that every element of $Sp(n)$ has determinant 1.

Proof. Let $U \in Sp(n)$. Note that the matrix U represents a linear transformation $L_u : \mathbb{C}^{2n} \rightarrow \mathbb{C}^{2n}$. By Theorem 1.6, there exists an orthonormal basis $u_1, \dots, u_n, v_1, \dots, v_n$ for $\mathbb{C}^{2n}$ such that

1. $Ju_j = v_j$

2. for some $\theta_1, \dots, \theta_n \in \mathbb{R}$,

$$Uu_j = e^{i\theta_j}u_j$$

$$Uv_j = e^{-i\theta_j}v_j$$

3. and we have

$$\omega(u_j, u_k) = \omega(v_j, v_k) = 0$$

$$\omega(u_j, v_k) = \delta_{jk}$$

The second point is the most important. Notice that it implies $e^{i\theta_j}$ are eigenvalues of U with eigenvectors u_j , and $e^{-i\theta_j}$ are eigenvalues of U with eigenvectors v_j .

Denote $B = \{u_1, \dots, u_n, v_1, \dots, v_n\}$. B is a basis for $\mathbb{C}^{2n}$ consisting entirely of eigenvectors of U , so U is diagonalizable. Thus there exists an invertible matrix $Q \in M_{2n}(\mathbb{C})$ such that

$$QUQ^{-1} = \text{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}, e^{-i\theta_1}, \dots, e^{-i\theta_n})$$

and so

$$U = Q^{-1} \text{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}, e^{-i\theta_1}, \dots, e^{-i\theta_n})Q$$

Now, we make a slight modification and define

$$U(t) = Q^{-1} \text{diag}(e^{i(1-t)\theta_1}, \dots, e^{i(1-t)\theta_n}, e^{-i(1-t)\theta_1}, \dots, e^{-i(1-t)\theta_n})Q$$

for $0 \leq t \leq 1$. $U(t)$ is continuous, $U(0) = U$, and $U(1) = Q^{-1}IQ = I$. By applying the converse of Theorem 1.6 to $U(t)$, it is clear that $U(t) \in Sp(n)$ for all $0 \leq t \leq 1$.

Hence $Sp(n)$ is connected.

For any $U \in Sp(n)$,

$$\begin{aligned} \det(U) &= \det(Q^{-1} \text{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}, e^{-i\theta_1}, \dots, e^{-i\theta_n})Q) \\ &= \det(Q^{-1}) \det(\text{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}, e^{-i\theta_1}, \dots, e^{-i\theta_n})) \det(Q) \\ &= \frac{1}{\det(Q)} e^{i\theta_1} \dots e^{i\theta_n} e^{-i\theta_1} \dots e^{-i\theta_n} \det(Q) \\ &= 1 \end{aligned}$$

□

Exercise 1.9 Suppose a is an irrational real number. Show that the set E_a of numbers of the form $e^{2\pi i n a}$, $n \in \mathbb{Z}$, is dense in the unit circle S^1 .

Proof. We follow the given hint, and divide S^1 into N equally sized bins of length $\frac{2\pi}{N}$. E_a has an infinite number of elements, and N may be large but finite. There must be at least one bin that has an infinite number of elements of E_a (otherwise, if every bin has finite number of elements and N is finite, then E_a would be finite).

Now, choose a bin with infinitely many elements of E_a . In this bin, it is possible to choose two distinct elements of E_a of the form $e^{2\pi i n a}$ and $e^{2\pi i m a}$, with $n, m \in \mathbb{Z}$, $n \neq m$.

We use the fact that E_a is a subgroup of S^1 . $e^{2\pi i n a}, e^{2\pi i m a} \in E_a$, so the inverse $e^{-2\pi i m a} \in E_a$, so $e^{2\pi i (n-m)a} \in E_a$, and so $(e^{2\pi i (n-m)a})^k = e^{2k\pi i (n-m)a} \in E_a$ for any $k \in \mathbb{Z}$.

By taking N to be large, we can force $e^{2\pi i n a}$ and $e^{2\pi i m a}$ to be arbitrarily close to each other on S^1 . Then $e^{2k\pi i (n-m)a}$ is a rotation by an arbitrarily small angle. Putting it to the power of k allows us to approximate any point on S^1 by an element of E_a .

More formally, let $e^{i\theta} \in S^1$ for any $\theta \in \mathbb{R}$.

$$\begin{aligned} |e^{i\theta} - e^{2k\pi i (n-m)a}| &= \left| e^{2k\pi i (n-m)a} \left(\frac{e^{i\theta}}{e^{2k\pi i (n-m)a}} - 1 \right) \right| \\ &= \left| \frac{e^{i\theta}}{e^{2k\pi i (n-m)a}} - 1 \right| \\ &= |e^{i(\theta - 2k\pi(n-m)a)} - 1| \\ &= |e^{i(\theta - 2k\pi(n-m)a)} - e^0| \end{aligned}$$

By continuity, $\forall \varepsilon > 0$, $\exists \delta > 0$ such that $|\theta - 2k\pi(n-m)a| < \delta \Rightarrow |e^{i(\theta - 2k\pi(n-m)a)} - e^0| < \varepsilon$. So $\forall \varepsilon > 0$, we can choose N large, and a suitable k such that

$$|e^{i\theta} - e^{2k\pi i (n-m)a}| < \varepsilon$$

Hence E_a is dense in S^1 . □

2 The Matrix Exponential

2.1 The Exponential of a Matrix

If X is a $n \times n$ matrix, we define the **exponential** of X , denoted e^X , by the power series

$$e^X = \sum_{m=0}^{\infty} \frac{X^m}{m!} \quad (2.1)$$

where X^0 is the identity matrix I and X^m is the repeated matrix product of X with itself.

Proposition 2.1. *The series 2.1 converges for all $X \in M_n(\mathbb{C})$ and e^X is a continuous function of X .*

Definition 2.2. *For any $X \in M_n(\mathbb{C})$, we define the **Hilbert-Schmidt** norm of X ,*

$$\|X\| = \left(\sum_{j,k=1}^n |X_{jk}|^2 \right)^{1/2}$$

The Hilbert-Schmidt norm may be computed in a basis-independent way as

$$\|X\| = (\text{tr}(X^* X))^{1/2}$$

This norm satisfies

$$\|X + Y\| \leq \|X\| + \|Y\|$$

$$\|XY\| \leq \|X\| \|Y\|$$

for all $X, Y \in M_n(\mathbb{C})$.

Proposition 2.3. *Let X and Y be arbitrary $n \times n$ matrices. Then we have the following:*

1. $e^0 = I$
2. $(e^X)^* = e^{X^*}$
3. e^X is invertible and $(e^X)^{-1} = e^{-X}$
4. $e^{(\alpha+\beta)X} = e^{\alpha X} e^{\beta X}$ for all α and β in $\mathbb{C}$
5. If $XY = YX$, then $e^{X+Y} = e^X e^Y = e^Y e^X$
6. If $C \in GL(N, \mathbb{C})$, then $e^{CXC^{-1}} = Ce^XC^{-1}$

Proposition 2.4. *Let X be a $n \times n$ complex matrix. Then e^{tX} is a smooth curve in $M_n(\mathbb{C})$ and*

$$\frac{d}{dt} e^{tX} = X e^{tX} = e^{tX} X$$

In particular,

$$\left. \frac{d}{dt} e^{tX} \right|_{t=0} = X$$

2.2 Computing the Exponential

The book explains some methods for computing the matrix exponential. If X is diagonalizable, then $X = CDC^{-1}$, and $e^X = Ce^DC^{-1}$. If X is nilpotent, ($X^k = 0$ for some k), then the power series of e^X terminates. Finally, every matrix X can be written uniquely as $X = S + N$ where S is diagonalizable, N is nilpotent, and $SN = NS$. Then $e^X = e^{S+N} = e^S e^N$.

2.3 The Matrix Logarithm

We want to define a matrix logarithm, which should be an inverse function to the matrix exponential. The simplest way is by a power series. Recall that in the case of complex numbers,

Lemma 2.5. *The function*

$$\log z = \sum_{m=1}^{\infty} (-1)^{m+1} \frac{(z-1)^m}{m}$$

is defined and holomorphic in a circle of radius 1 about $z = 1$. For all z with $|z - 1| < 1$,

$$e^{\log z} = z$$

For all u with $|u| < \log 2$, we have $|e^u - 1| < 1$ and

$$\log e^u = u$$

Definition 2.6. *For an $n \times n$ matrix A , define $\log A$ by*

$$\log A = \sum_{m=1}^{\infty} (-1)^{m+1} \frac{(A - I)^m}{m}$$

whenever the series converges.

Theorem 2.7. *The function*

$$\log A = \sum_{m=1}^{\infty} (-1)^{m+1} \frac{(A - I)^m}{m}$$

is defined and continuous on the set of all $n \times n$ complex matrices A with $\|A - I\| < 1$. For all $A \in M_n(\mathbb{C})$ with $\|A - I\| < 1$,

$$e^{\log A} = A$$

For all $X \in M_n(\mathbb{C})$ with $\|X\| < \log 2$, $\|e^X - I\| < 1$ and

$$\log e^X = X$$

Notice from the last sentence that it is not true that $\log e^X = X$ whenever the series for the logarithm converges.

Proposition 2.8. *There exists a constant c such that for all $n \times n$ matrices B with $\|B\| < 1/2$, we have*

$$\|\log(I + B) - B\| \leq c\|B\|^2$$

More concisely, the proposition can be stated as

$$\log(I + B) = B + O(\|B\|^2)$$

Theorem 2.9. *Every invertible $n \times n$ matrix can be expressed as e^X for some $X \in M_n(\mathbb{C})$.*

2.4 Further Properties of the Exponential

Theorem 2.10 (Lie Product Formula). *For all $X, Y \in M_n(\mathbb{C})$, we have*

$$e^{X+Y} = \lim_{m \rightarrow \infty} \left(e^{\frac{X}{m}} e^{\frac{Y}{m}} \right)^m$$

Theorem 2.11. *For any $X \in M_n(\mathbb{C})$, we have*

$$\det(e^X) = e^{\text{tr}(X)}$$

Definition 2.12. *A function $A : \mathbb{R} \rightarrow GL(n, \mathbb{C})$ is called a **one-parameter subgroup** of $GL(n, \mathbb{C})$ if*

1. A is continuous,
2. $A(0) = I$,
3. $A(t+s) = A(t)A(s)$ for all $t, s \in \mathbb{R}$.

Theorem 2.13 (One-Parameter Subgroups). *If $A(\cdot)$ is a one-parameter subgroup of $GL(n, \mathbb{C})$, there exists a unique $n \times n$ complex matrix X such that*

$$A(t) = e^{tX}$$

Lemma 2.14. *Fix some ε with $\varepsilon < \log 2$. Let $B_{\varepsilon/2}$ be the ball of radius $\varepsilon/2$ around the origin in $M_n(\mathbb{C})$, and let $U = \exp(B_{\varepsilon/2})$. Then every $B \in U$ has a unique square root C in U , given by $C = \exp(\frac{1}{2} \log B)$.*

Proposition 2.15. *The exponential map is an infinitely differentiable map of $M_n(\mathbb{C})$ into $M_n(\mathbb{C})$.*

2.5 The Polar Decomposition

The polar decomposition for a nonzero complex number z states that z can be written uniquely as $z = up$, where $|u| = 1$ and p is real and positive. Since p is real and positive, it can be written as $p = e^x$ for a unique real number x . This gives the polar decomposition for z ,

$$z = ue^x$$

with $x \in \mathbb{R}$ and $|u| = 1$.

We want to establish a similar polar decomposition for $GL(n, \mathbb{C})$, then for various subgroups. If P is a self-adjoint $n \times n$ matrix ($P^* = P$), we say that P is **positive** if $\langle v, Pv \rangle > 0$ for all nonzero $v \in \mathbb{C}^n$. A self-adjoint matrix P is positive if and only if all the eigenvalues of P are positive.

Theorem 2.16. *Polar decomposition for $GL(n, \mathbb{C})$.*

1. Every $A \in GL(n, \mathbb{C})$ can be written uniquely in the form

$$A = UP$$

where U is unitary and P is self-adjoint and positive.

2. Every self-adjoint positive matrix P can be written uniquely in the form

$$P = e^X$$

with X self-adjoint. Conversely, if X is self-adjoint, then e^X is self-adjoint and positive.

3. If we decompose each $A \in GL(n, \mathbb{C})$ (uniquely) as

$$A = Ue^X$$

with U unitary and X self-adjoint, then U and X depend continuously on A .

Lemma 2.17. If Q is a self-adjoint, positive matrix, then Q has a unique self-adjoint, positive square root.

Todo: write proof of theorem

Proposition 2.18. Polar decomposition for various subgroups of $GL(n, \mathbb{C})$.

1. Every $A \in GL(n, \mathbb{R})$ can be written uniquely as

$$A = Re^X$$

where $R \in O(n)$ and X is real and symmetric.

2. Every $A \in SL(n, \mathbb{C})$ can be written uniquely as

$$A = Ue^X$$

where $U \in SU(n)$ and X is self-adjoint with trace zero.

3. Every $A \in SL(n, \mathbb{R})$ can be written uniquely as

$$A = Re^X$$

where $R \in SO(n)$ and X is real and symmetric with trace zero.

2.6 Exercises

Exercise 2.6 Show that every 2×2 matrix X with $\text{tr}(X) = 0$ satisfies

$$X^2 = -\det(X)I$$

If X is 2×2 with trace zero, show by direct calculation using the power series for the exponential that

$$e^X = \cos(\sqrt{\det X})I + \frac{\sin \sqrt{\det X}}{\sqrt{\det X}}X$$

where $\sqrt{\det X}$ is either of the two (possibly complex) square roots of $\det X$. Use this to give an alternative computation of the exponential e^{X_1} in Example 2.5.

Proof. Let X be a 2×2 matrix with $\text{tr}(X) = 0$, taking the form

$$X = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}$$

The determinant is

$$\det(X) = -a^2 - bc$$

Hence, we have

$$X^2 = \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \begin{pmatrix} a & b \\ c & -a \end{pmatrix} = \begin{pmatrix} a^2 + bc & 0 \\ 0 & a^2 + bc \end{pmatrix} = (a^2 + bc)I = -\det(X)I$$

which proves the required statement.

The power series of the matrix exponential is

$$e^X = \sum_{n=0}^{\infty} \frac{X^n}{n!} = I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \frac{X^4}{4!} + \cdots$$

Using the result we just proved, $X^2 = -\det(X)I$, we can generalize and say that

$$X^{2n} = (-1)^n \det(X)^n I$$

$$X^{2n+1} = X^{2n} X = (-1)^n \det(X)^n X$$

The power series of the matrix exponential is absolutely convergent, so we can rearrange terms without worrying about the series converging to some other limit (source).

$$\begin{aligned} e^X &= I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \frac{X^4}{4!} + \cdots \\ &= \left(I + \frac{X^2}{2!} + \frac{X^4}{4!} + \cdots\right) + \left(X + \frac{X^3}{3!} + \frac{X^5}{5!} + \cdots\right) \\ &= \left(\sum_{n=0}^{\infty} \frac{(-1)^n \det(X)^n}{(2n)!}\right) I + \left(\sum_{n=0}^{\infty} \frac{(-1)^n \det(X)^n}{(2n+1)!}\right) X \\ &= \left(\sum_{n=0}^{\infty} \frac{(-1)^n \sqrt{\det(X)}^{2n}}{(2n)!}\right) I + \left(\sum_{n=0}^{\infty} \frac{(-1)^n \sqrt{\det(X)}^{2n+1}}{\sqrt{\det(X)}(2n+1)!}\right) X \\ &= \cos(\sqrt{\det X}) I + \frac{\sin \sqrt{\det X}}{\sqrt{\det X}} X \end{aligned}$$

where the last equality is from the power series of $\cos$ and $\sin$. This proves the desired identity.

In Example 2.5, the matrix X_1 is

$$X_1 = \begin{pmatrix} 0 & -a \\ a & 0 \end{pmatrix}$$

which is a 2×2 matrix with trace zero. Its determinant is $\det X_1 = a^2$. We apply the above formula to compute its matrix exponential.

$$\begin{aligned} e^{X_1} &= \cos(\sqrt{a^2}) I + \frac{\sin(\sqrt{a^2})}{\sqrt{a^2}} X_1 \\ &= \begin{pmatrix} \cos(\sqrt{a^2}) & 0 \\ 0 & \cos(\sqrt{a^2}) \end{pmatrix} + \begin{pmatrix} 0 & -a \frac{\sin(\sqrt{a^2})}{\sqrt{a^2}} \\ a \frac{\sin(\sqrt{a^2})}{\sqrt{a^2}} & 0 \end{pmatrix} = \begin{pmatrix} \cos a & -\sin a \\ \sin a & \cos a \end{pmatrix} \end{aligned}$$

This is the same result as in Example 2.5. Note that $\sqrt{\det X_1} = \sqrt{a^2}$ can be taken to be either $+a$ or $-a$. The final result is the same. $\square$